Advanced Fixed Income and Credit

Group Assignment

Part I: Deutsche Bank — Finding Relative-Value Trades (HBS 205-059)
Part II: Python Labs — US Treasury Yield Curve Reaction to Trump Tariff Threats

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1 Part I — The Deutsche Bank Case Study

Setup and conventions

Settlement date: **August 15, 2003**. Exhibit 1 lists 52 Treasury coupon bonds whose maturities fall on successive semi-annual dates (Feb 15 / Aug 15) from 2004 to 2029 with no gaps, i.e. exactly one bond at each of the nodes $t = 0.5, 1.0, \ldots, 26.0$ years. This allows the classic *successive substitution* bootstrap: because each bond introduces exactly one new unknown discount factor, we solve sequentially from the shortest to the longest maturity.

Compounding convention. Exhibit 4's footnote states explicitly that Deutsche Bank's own model yields are *Bond-Equivalent Yields* (BEY), i.e. twice the semi-annual compounded yield. Since U.S. Treasuries pay semi-annual coupons, we derive our own zero-coupon yields in the identical BEY convention throughout, so that every comparison against the DB model in Q1.2 is apples-to-apples (a continuously-compounded convention would have introduced a small but systematic bias in one direction).

$$\text{BEY} = 2\left(D^{-1/k} - 1\right), \qquad D = \text{discount factor to period } k \text{ (6-month steps)}.$$

Q1.1 — Bootstrapping the discount curve

Methodology. For bond k (maturing at period k, coupon c per annum, price P_k):

$$P_k = \frac{c}{2}\sum_{j=1}^{k-1} D_j + \left(\frac{c}{2} + 100\right) D_k \quad \Longrightarrow \quad D_k = \frac{P_k - \frac{c}{2}\sum_{j=1}^{k-1} D_j}{\frac{c}{2} + 100}.$$

Solved in order of increasing maturity, using only already-known D_j, $j < k$.

Validation. Our bootstrapped curve matches Deutsche Bank's own Exhibit 4 model yields *exactly* at 2Y (1.873%) and 10Y (4.711%) — precisely the maturities the case identifies as DB's own calibration points for its three-factor model. This is strong internal validation that both the bootstrap algorithm and the BEY convention choice are correct.

Maturity (y)	Discount Factor	Zero Yield (BEY)
0.5	0.99561	0.882%
1.0	0.98818	1.193%
2.0	0.96341	1.873%
5.0	0.84156	3.480%
10.0	0.62775	4.711%
15.0	0.42935	5.717%
20.0	0.30947	5.951%
25.0	0.23299	5.913%
26.0	0.22024	5.905%

Table 1: Bootstrapped zero-coupon curve, selected maturities (full 52-point curve in `q1_bootstrapped_curve.csv`).

The curve shape is the standard concave, upward-sloping Treasury curve of the period, flattening from about 15Y onward and showing a very slight downward drift between 20Y and 26Y.

Q1.2 — Comparison with Deutsche Bank's model, trading signals

Logic. Yield and price move in opposite directions. If our bootstrapped (market-implied) yield is *higher* than DB's model yield, the bond's market price is below the model's "fair" price ⇒ the bond is **cheap** ⇒ **BUY**. If the market yield is *lower*, the bond is **rich** ⇒ **SELL**.

Pattern. The belly of the curve (3–9 years) is uniformly *cheap* relative to the DB model, by 1–6 bps, suggesting a BUY candidate basket in that region. The very short end (1Y) and the long

Maturity (y)	DB Model Yield	Market Yield (Q1.1)	Diff (bps)	Signal
1	1.2443%	1.1930%	−5.13	SELL (rich)
2	1.8727%	1.8727%	+0.00	fair (DB calibration point)
3	2.4110%	2.4711%	+6.01	**BUY (cheap)**
4	2.9665%	3.0084%	+4.19	**BUY (cheap)**
5	3.4454%	3.4799%	+3.45	**BUY (cheap)**
6	3.8557%	3.8817%	+2.60	**BUY (cheap)**
7	4.1996%	4.2203%	+2.07	**BUY (cheap)**
8	4.4677%	4.4865%	+1.88	**BUY (cheap)**
9	4.6528%	4.6640%	+1.12	**BUY (cheap)**
10	4.7107%	4.7107%	−0.00	fair (DB calibration point)
15	5.7160%	5.7167%	+0.07	fair
20	5.9517%	5.9513%	−0.04	fair
25	5.9315%	5.9128%	−1.87	SELL (rich)

Table 2: Market vs. DB model comparison and trading signals.

end (25Y) look *rich*. The 2Y, 10Y, 15Y and 20Y points are essentially fair by construction or coincidence (2Y and 10Y are DB's own calibration anchors).

Q1.3 — Is this risk-free arbitrage?

No. Although the strategy is directional and model-based, it is **not** a riskless arbitrage, for several reasons:

- **Model risk.** The "mispricing" is only a mispricing relative to Deutsche Bank's own three-factor curve model. If the model is mis-specified (wrong number of factors, wrong functional form, stale calibration), the signal is simply wrong — there is no model-independent proof that the belly of the curve is "truly" cheap.
- **Convergence/timing risk.** Even if the model is correctly specified on average, there is no guarantee *when*, or even *whether*, market yields converge back toward the model yields within the trader's holding horizon. The position carries interim mark-to-market risk.
- **No riskless, capital-structure-free payoff.** A true arbitrage requires a replicating, self-financing portfolio with a knowable, riskless terminal payoff (e.g. a textbook cash-and-carry). Here the "trade" is an outright directional bet on convergence, funded at some financing rate, with basis risk between the position and any hedge.
- **Curve risk while waiting.** Between trade inception and convergence, the *whole* curve can move (parallel shifts, twists), and unless the position is duration/key-rate hedged, those moves swamp the small model-implied edge of a few basis points.

In short, this is a **relative-value** trade: a model-informed bet with positive expected value under the trader's own beliefs about the model's validity, not a riskless arbitrage in the classical sense.

Q2.1 — Cubic spline of the zero-coupon curve

Methodology. Natural cubic spline (zero second derivative at the boundary knots) through the five knots specified (2, 5, 10, 15, 20 years), fit directly on the Q1.1 bootstrapped BEY yields. Outside $[2, 20]$y the spline is extrapolated using its boundary slope.

Fit quality. Within the knot range $[2, 20]$y: mean residual $= -5.50$ bps, std $= 8.28$ bps, max $|\text{residual}| = 29.91$ bps. The residuals are *not* randomly scattered: the largest errors cluster sharply between 7.5 and 13 years (up to -30 bps at 11.5Y), exactly the maturity window in which Exhibit 1 contains several legacy, high-coupon bonds issued in the early 1980s (coupons of 13.25%, 12.5%,

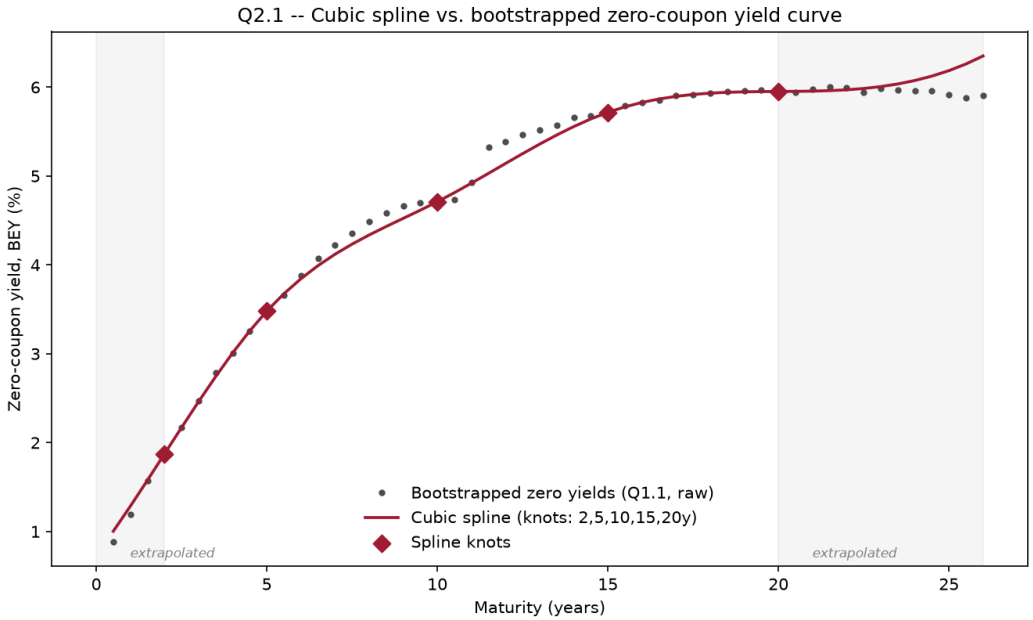


Figure 1: Cubic spline (red) vs. raw bootstrapped zero yields (grey dots). Shaded regions are extrapolated.

11.25%, 10.625%). These bonds plausibly carry idiosyncratic pricing noise (age, liquidity, possible embedded optionality/callability common to that vintage), which the raw bootstrap faithfully reproduces pointwise but which the smooth spline, by construction, averages away.

Q2.2 — Does using the spline change the Q1.2 conclusions?

We repeat the Q1.2 comparison using the **spline-implied** yield (instead of the raw bootstrap) at every maturity used in the original table, since the spline value differs from the raw bootstrap except exactly at the five knots (2,5,10,15,20).

Maturity (y)	DB Model	Spline Yield	Diff (bps)	Spline Signal	*Changed vs. Q1.2?*
1	1.2443%	1.2846%	+4.03	BUY	**YES** (was SELL)
2	1.8727%	1.8727%	+0.02	fair	no (knot)
3	2.4110%	2.4608%	+4.98	BUY	no
4	2.9665%	3.0096%	+4.31	BUY	no
5	3.4454%	3.4799%	+3.45	BUY	no (knot)
6	3.8557%	3.8439%	−1.18	SELL	**YES** (was BUY)
7	4.1996%	4.1195%	−8.01	SELL	**YES** (was BUY)
8	4.4677%	4.3363%	−13.14	SELL	**YES** (was BUY)
9	4.6528%	4.5235%	−12.93	SELL	**YES** (was BUY)
10	4.7107%	4.7107%	−0.00	fair	no (knot)
15	5.7160%	5.7167%	+0.07	fair	no (knot)
20	5.9517%	5.9513%	−0.04	fair	no (knot)
25	5.9315%	6.1859%	+25.44	BUY	**YES** (was SELL)

Table 3: Q1.2 signals recomputed off the Q2.1 spline curve.

This matters a great deal. Six of the thirteen signals (1, 6, 7, 8, 9, 25 years) *flip sign* once the spline replaces the raw bootstrap. Two distinct mechanisms are at work:

1. **Extrapolation risk (1Y and 25Y).** These two points lie *outside* the spline's knot range $[2, 20]$y. The natural spline's boundary-slope extrapolation is a pure modelling artifact with no

claim to describing the true curve shape beyond the data used to fit it, so any signal built on it is unreliable almost by definition.

2. **Belly-of-curve smoothing (6–9Y).** This is the more substantive finding. The original Q1.2 "cheap" signal in 6–9Y was driven, at least in part, by the *same* idiosyncratic legacy-bond noise identified in Q2.1's residual analysis. Once the spline smooths through that local bump, the belly of the curve actually looks *rich*, not cheap, relative to the DB model.

Conclusion. The trading conclusion from Q1.2 is *not robust* to the choice of curve-fitting methodology in exactly the region (6–9Y) where the raw bootstrap is most likely contaminated by non-fundamental, bond-specific noise. A prudent trader would treat the raw-bootstrap BUY signal in 6–9Y with real skepticism, and would want to understand *why* those bonds are mispriced (liquidity? embedded option? stale quote?) before trading on it, rather than assuming genuine relative value.

Q3.1 — Nelson-Siegel-Svensson fit

Methodology. Jointly estimating $(\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ by unconstrained nonlinear least squares proved numerically fragile: this curve is concave with a single mild hump and no second hump, so β_0 (level) and β_2 (curvature) become highly collinear once τ_1 is pushed above ∼10 years (the curvature loading then barely decays within our 26-year range). Different, equally-well-fitting local optima emerged from different starting points.

We instead use the standard **Diebold–Li two-step procedure**: for each candidate (τ_1, τ_2) on a fine grid, the loadings become fixed basis functions of maturity, so the betas enter *linearly* and are solved by OLS — a well-posed, unique, convex problem for every candidate τ. We then pick the (τ_1, τ_2) minimizing total RSS.

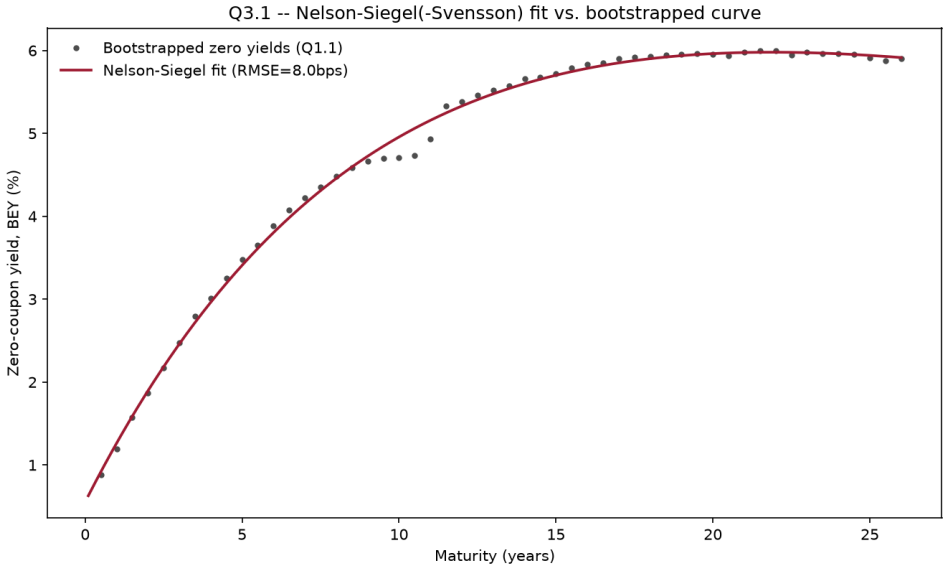


Figure 2: Nelson-Siegel fit vs. the Q1.1 bootstrapped curve.

Model selection. The plain 3-factor Nelson-Siegel achieves RMSE = 7.96 bps ($\beta_0 = 0.695\%$, $\beta_1 = -0.140\%$, $\beta_2 = 17.926\%$, $\tau_1 = 12.08$y). The full 4-factor Svensson only improves RMSE to 7.28 bps (an improvement of 0.68 bps), below our pre-set materiality threshold of 1.5 bps. We therefore adopt the **simpler, better-identified 3-factor Nelson-Siegel model**: the data show no evidence of a genuine second hump, so the extra parameter and its attendant identification problems are not worth the marginal fit improvement.

Parameter	Value
β_0 (level)	0.695%
β_1 (slope)	−0.140%
β_2 (curvature)	17.926%
τ_1	12.081 years
RMSE	7.96 bps

Table 4: Final adopted Nelson-Siegel parameters.

Q3.2 — Factor loadings and economic drivers

The contribution decomposition reveals a striking feature: the **curvature** factor ($\beta_2 = 17.9\%$) does almost all of the curve-shaping work, rising smoothly from 0 at $m = 0$ to its asymptote around 20 years, while the level contribution is small and essentially flat (0.695%) and the slope contribution is tiny and nearly flat (−0.140%). This is counter-intuitive at first glance — in textbook Nelson-Siegel intuition, "level" should dominate a curve of this overall magnitude — but it is a direct, mechanical consequence of τ_1 being large (12.08 years) relative to our 26-year maturity range: the curvature loading function $\left[\frac{1-e^{-m/\tau_1}}{m/\tau_1} - e^{-m/\tau_1}\right]$ simply has not decayed back to zero by $m = 26$, so it is statistically *behaving like* a second, slowly-varying level/slope factor over the range of our data, and the OLS step naturally loads most of the explanatory power onto it rather than onto β_0 precisely because β_0's loading is flat and uninformative about curve *shape*, whereas the curvature loading can trace the whole concave shape on its own.

Economically, this echoes Deutsche Bank's own Exhibit 3 three-factor model description: a persistent, long-horizon "long-rate"/level-type factor that evolves slowly is naturally captured by a Nelson-Siegel factor whose decay parameter is large relative to the fitting window, rather than by the model's nominal level factor, which by construction carries no shape information at all. In other words, when the sample's maturity range is not much longer than τ_1, Nelson-Siegel's "curvature" factor and a slow-moving macro "long rate" factor become difficult to tell apart — exactly the level/curvature identifiability issue we had already flagged during estimation (Q3.1).

Q4 — Answer summary

Q4.1. A +50bps bump of the 10Y knot with the natural spline re-fitted raises the 10Y zero yield by exactly 50bps and leaves the other four knots unchanged, but the shock is *not* local: the curve ripples (+22.1bps at 7Y, −8.4bps at 17Y, −11.5bps at 26Y).

Q4.2. KRD_{10Y} = 2.21 years (ΔP = −\$10,813.14, i.e. −1.08% of NAV; DV01 = \$221.0 central, \$216.3 one-sided).

Q4.3. The 52 individual zero-yield KRDs sum to 7.981 years, exactly the portfolio's effective duration; the aggregated key-rate profile is 0.37/1.54/2.21/0.81/3.06 years at 2/5/10/15/20Y.

Q4.1 — Shocking the 10Y key rate

Methodology. The Q2.1 curve is a natural cubic spline through the bootstrapped zero yields at the knots 2, 5, 10, 15 and 20y, which are exactly the assignment's key rates. We therefore implement "shock the 10-year rate by 50bps" literally on that model: we add +50bps to the 10Y knot value, *re-fit* the natural spline, and define the curve shock as

$$\Delta y(t) = S_{\text{bumped}}(t) - S_{\text{base}}(t), \qquad t = 0.5, 1.0, \ldots, 26\text{y}.$$

Because spline interpolation is linear in the knot values, $\Delta y(t)$ does not depend on the level of the curve, and the five key-rate shocks sum *exactly* to a +50bps parallel shift at every maturity (verified in the code). For comparison, we also run the textbook Ho (1992) triangular ("tent") shock: full size at its own knot, decaying linearly to zero at the adjacent knots, and held flat beyond the outer knots.

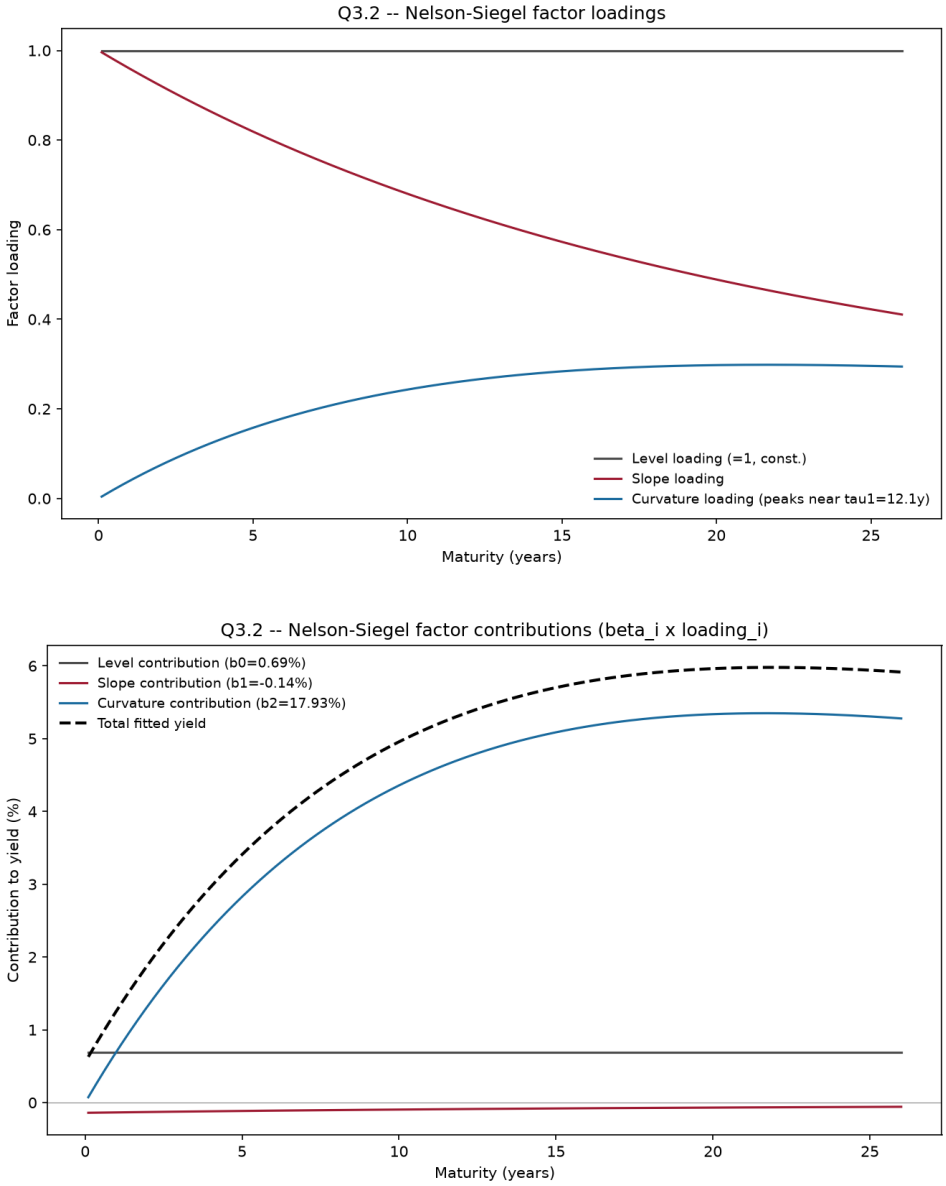


Figure 3: Top: factor loadings. Bottom: factor contributions ($\beta_i \times$ loading$_i$).

Impact on the curve. The shock is +50bps at 10Y and, by construction, zero at the other knots (2, 5, 15, 20y). However, unlike the tent, the spline response is **not local**:

- Between 5Y and 15Y it is *wider* than the tent: +22.1bps at 7y and +37.0bps at 12y, versus +20 and +30bps for the linear ramp.
- Outside $[5, 15]$y the cubic *ripples* with alternating signs: about −3 to −4bps between 3 and 4y, −8.4bps at 17y, and +8.4bps at 23y.
- In the **extrapolated** region beyond 20y the spline continues the last cubic polynomial — the `CubicSpline` implementation extends the end cubic, it does not hold the curve flat — so the ripple grows to −11.5bps at 26y. (A flat extension would put the 26Y yield at 5.95% instead of 6.35% and would set this last contribution to zero; we keep the model-consistent extension and quantify the sensitivity in Q4.3.) The right panel shows that this effect is much stronger for the outer knots: a 20Y bump moves the 26y yield by about +103bps, a 15Y bump moves it by about −44bps, and a 2Y bump moves the 0.5y yield by +79bps.

In short, a key-rate shock is only as local as the curve model it is applied to. The tent is local by design, while a cubic spline spreads a single-node shock across the whole curve, most strongly where the curve is extrapolated rather than interpolated.

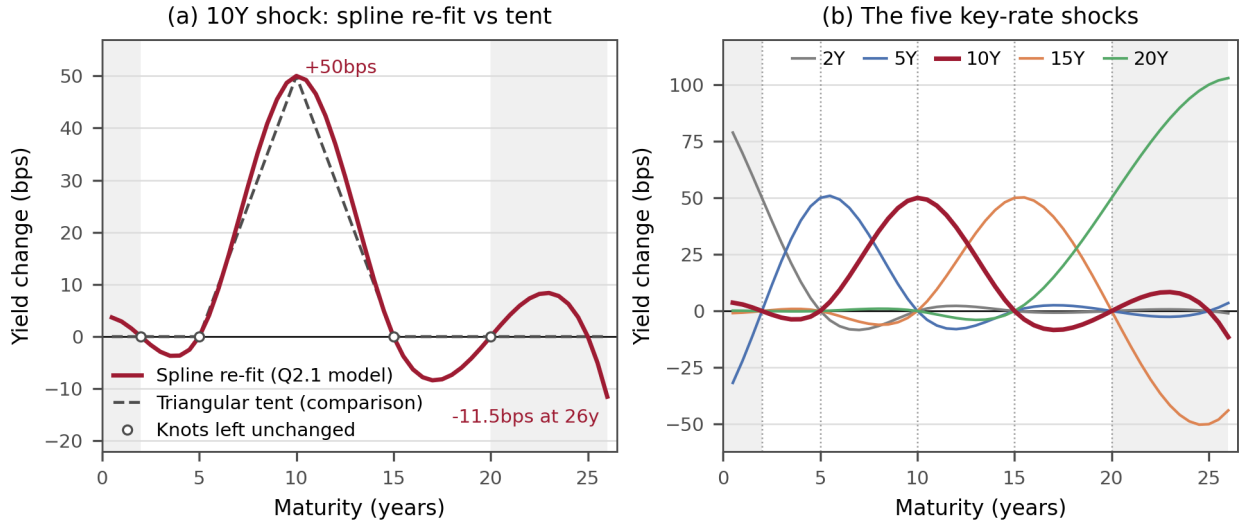


Figure 4: Left: the curve shock $\Delta y(t)$ from re-fitting the spline (red) vs. the triangular tent (dashed); hollow markers are the four knots left unchanged. Right: the five spline key-rate shocks. Shaded areas: extrapolated maturities (< 2y, > 20y).

Maturity (y)	2	3	5	7	10	12	15	17	20	23	26
Δy, spline re-fit (bps)	0.0	−2.9	0.0	+22.1	+50.0	+37.0	0.0	−8.4	0.0	+8.4	−11.5
Δy, tent (bps)	0.0	0.0	0.0	+20.0	+50.0	+30.0	0.0	0.0	0.0	0.0	0.0

Table 5: The +50bps 10Y shock at selected maturities: global spline re-fit vs. the local triangular tent. Δy is zero at 5Y and 15Y because the tent is a straight line between the knots; it is zero at 2, 5, 15 and 20Y for the re-fit because those knots are not moved.

Q4.2 — Portfolio KRD to the 10Y shock

Portfolio construction (assumption). "Equally invested" is read as an equal *dollar market value*: \$1,000,000/52 ≈ \$19,230.77 in each bond, i.e. a face amount of (dollar amount/price) × 100 per bond. The alternative reading, an equal *face* amount (\$19,230.77 of face in each bond), would give a \$1.19m market value and $\text{KRD}_{10Y} = 2.31$y: equal face overweights the cheap short bonds and underweights the expensive long ones, so it raises duration. We use equal market value, the standard reading of "equally invested", and report the alternative in the robustness table.

Approach.

1. Build the 52 × 52 cash-flow matrix (semi-annual coupons plus principal) and aggregate it into the portfolio cash flow at each semi-annual date.
2. *Base valuation* uses the Q1.1 bootstrapped zero curve (BEY, semi-annual compounding). It reprices every bond exactly, so the base NAV is exactly \$1,000,000. Pricing off the spline itself would give \$1,002,389, and the \$2,389 gap (the Q2.2 fitting residual) would contaminate the risk measure.
3. *Shocked valuation*: add the spline-implied $\Delta y(t)$ from Q4.1 to the bootstrapped yields and reprice. The shape of the shock therefore comes from the Q2.1 model, while the base level matches market prices.
4. Report the result as (i) the dollar P&L of the +50bps shock; (ii) the KRD in years, using a central difference to strip out convexity, $\text{KRD}_k = -\dfrac{P(+h) - P(-h)}{2\,P_0\,h}$ with $h = 50$bps; and (iii) the DV01 (\$ per 1bp), which is the quantity used to size hedges.

Result.

$\Delta P_{10Y} = \$989{,}186.86 - \$1{,}000{,}000.00 = -\$10{,}813.14$ (−1.08% of NAV), $\text{KRD}_{10Y} = 2.21$ years, DV01_1

The sign is as expected: a long bond portfolio loses value when yields rise. The \$221.0 is a *central-difference* DV01, DV01 $= -(P(+h)-P(-h))/2\times10^{-4}$; the *one-sided* +50bps P&L implies \$216.26 per bp, so $\Delta P \neq -50 \times$ DV01. The \$4.8 per bp gap is the convexity term, which a 50bp move is large enough to reveal; we report the central figure because it is the one used to size the hedge in Q4.3.

What actually moves the number (robustness). The KRD is essentially invariant to the pricing engine and materially sensitive to the *shock design*, which is where most of the dispersion between reasonable implementations comes from:

Variant	ΔP for +50bps	KRD_{10Y} (yrs)
Spline re-fit, bootstrapped base (**reported**)	−\$10,813	2.210
Spline re-fit, spline-priced base	−\$10,954	2.234
Triangular tent, bootstrapped base	−\$10,123	2.061
Triangular tent, spline-priced base	−\$10,232	2.078
Single-point 10Y bump (10Y zero yield only)	−\$1,169	0.240
Equal-face portfolio, bootstrapped base	−\$13,445	2.307

Every defensible convention puts the 10Y exposure between 2.0 and 2.4 years; the spread is methodological, not arithmetic. We report 2.21y because (i) Q4.1 asks for the impact on the curve *estimated in Q2.1*, which is the re-fitted spline, and (ii) pricing the base off the bootstrap keeps P_0 exactly at the \$1m actually invested, so the −\$2,389 Q2.2 fitting residual cannot leak into the risk measure. The 0.24y row shows the opposite extreme: applying the shock to the single 10Y zero yield only measures the partial duration of one cash-flow date, not a key-rate duration.

Q4.3 — Full key-rate duration profile

(a) Each of the 52 zero-coupon yields individually (the literal reading). "All maturities" is read as applying the same +50bps shock to each semi-annual zero yield (0.5y to 26y) *on its own*, leaving the other 51 unchanged. This version needs no curve model: only the discount factor of that single date moves, so to first order the KRD at date t is $t \times \text{PV}(\text{CF}_t)/\text{NAV}$ (i.e. $[t/(1+y_t/2)] \times \text{PV}(\text{CF}_t)/\text{NAV}$ in BEY terms).

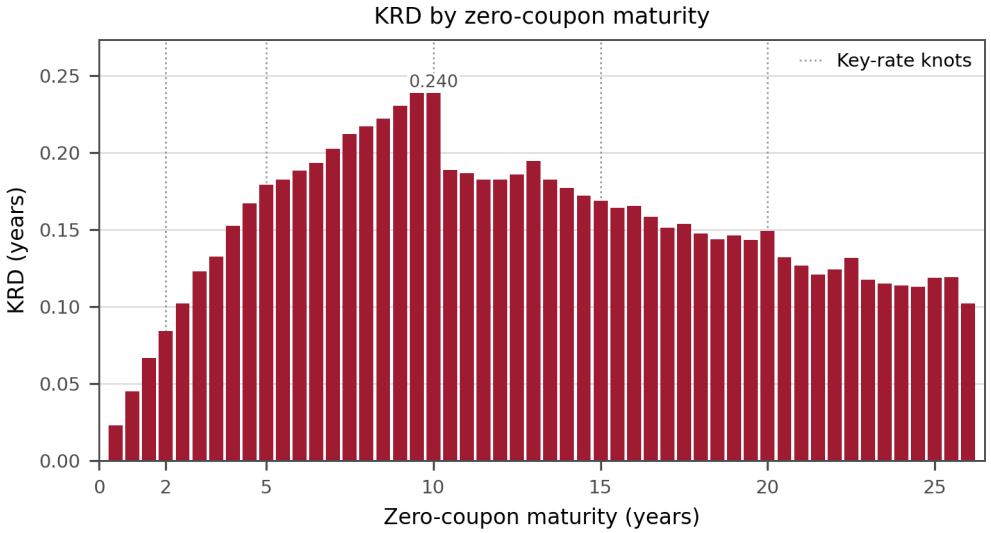


Figure 5: KRD to each individual zero-coupon yield (+50bps). The 52 bars sum to 7.981 years, the portfolio's effective duration.

Additivity check. The 52 KRDs sum to 7.9809 years, exactly the effective duration of a ±50bps parallel shift: a parallel shift is the sum of the 52 single-date shocks, so this check needs no curve model and is free of cross terms.

Shape of the profile. The exposure at date t is roughly $t\times$(PV of the portfolio cash flow at t)/NAV. It rises with t to a peak of 0.24y at 10y, then *drops abruptly* at 10.5y. Bonds maturing from Feb-2014 onwards carry high coupons (5.25–13.25%) and mostly trade far above par (up to 174), so an equal *dollar* allocation buys less face — hence less principal — at those dates. Beyond 10.5y the exposure declines slowly as discounting erodes the PV of distant cash flows.

(b) The five key rates (aggregated view). We apply the same +50bps re-fit shock to each knot in turn.

	Spline re-fit (Q2.1 model)			Tent (comparison)	
Key rate	P&L ($)	KRD (yrs)	DV01 ($)	P&L ($)	KRD (yrs)
2Y	−1,820	0.367	36.7	−2,640	0.531
5Y	−7,594	1.541	154.1	−7,156	1.446
10Y	−10,813	2.210	221.0	−10,123	2.061
15Y	−3,496	0.810	81.0	−8,236	1.690
20Y	−13,980	3.064	306.4	−10,717	2.252
Sum	−37,704	7.992	799.2	−38,872	7.979

Table 6: Key-rate exposures of the $1M equally-weighted portfolio (+50bps shock at each knot). The parallel-shift effective duration is 7.981 years. KRD and DV01 are central differences; the P&L column is one-sided, so P&L $\neq -50\times$DV01.

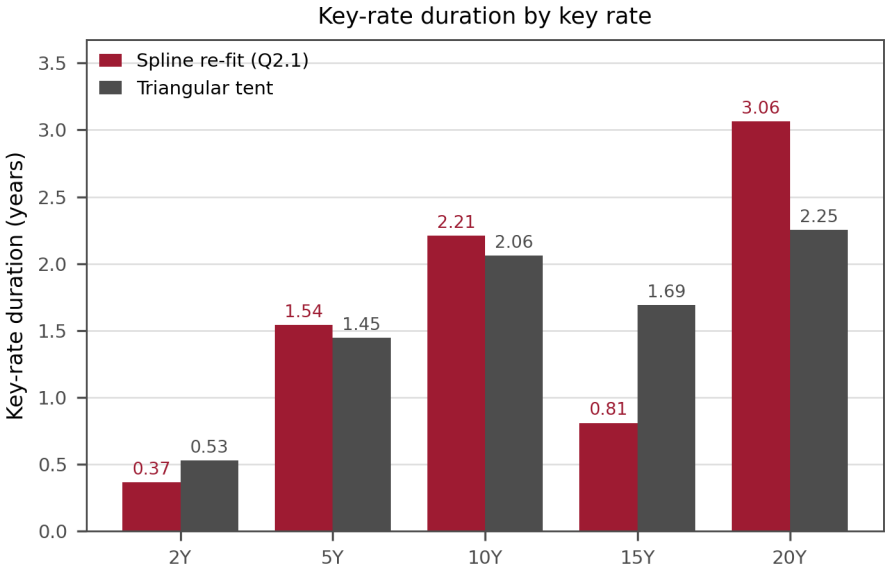


Figure 6: Key-rate duration profile (years): spline re-fit vs. triangular tent.

Consistency check. Both families of shocks sum to a parallel shift, so the KRDs should add up to the effective duration. They do: 7.992y (spline) and 7.979y (tent), versus 7.981y for a ±50bps parallel shift. Summing the *one-sided* dollar P&Ls gives −$37,704 versus −$38,594 for the parallel shock. The gap comes from convexity and cross terms, which are largest for the spline because its extrapolated shocks reach about 100bps. This is why we report central-difference KRDs.

Robustness: the 15Y/20Y split is an extrapolation effect. The Q2.1 spline is pinned by data only up to 20Y; beyond that, the end cubic amplifies the effect of a 50bps bump at 20Y to +103bps at 26Y, and the effect of a 50bps bump at 15Y to −50bps at 24.5Y. That is what moves duration between the two buckets:

	2Y	5Y	10Y	15Y	20Y	Sum
End cubic (reported)	0.367	1.541	2.210	0.810	3.064	7.992
Held flat beyond 20Y	0.180	1.619	2.109	1.850	0.652	6.411

Holding the curve flat would move the split to 1.85y (15Y) and 0.65y (20Y), but the five KRDs would then sum to 6.41y instead of the portfolio's 7.98y effective duration, because the shocks would no longer sum to a parallel shift: the cash flows beyond 20Y, about 1.57y of duration, would carry no key-rate exposure at all. We keep the model-consistent extension — it is the only one under which the key-rate decomposition is complete — and present the 15Y/20Y split as model risk rather than as a tradable exposure.

Interpretation.

- All exposures are positive durations (negative P&L), as expected for a long-only portfolio.
- With the tent, the profile is smooth (0.53 / 1.45 / 2.06 / 1.69 / 2.25y) and follows the cash-flow intuition; the 20Y bucket is the largest because it also absorbs every maturity from 20 to 26y.
- With the spline re-fit, the ripples of Q4.1 redistribute risk. The 2Y KRD falls to 0.37y (vs. 0.53y): the 2Y bump pushes 5–10y yields *down* by up to 8.4bps, worth +\$1,063 on the 5–10y cash flows, which more than offsets the extra −\$219 caused by its larger short-end shock (+79bps at 0.5y vs. +50bps for the tent). The 15Y/20Y split moves as explained above. The total is unchanged, which is the point.
- **Practical implication.** The choice of curve model is itself a source of hedging risk. A hedge sized on spline KRDs would put far more weight on 20Y and far less on 15Y than a hedge sized on tent KRDs, even though both imply the same total duration.

Illustrative KRD hedge (beyond the question). Neutralising the five spline key-rate DV01s with the five bonds maturing exactly at the key dates requires short \$93.6k (Aug-2005), short \$199.5k (Aug-2008), short \$211.5k (Aug-2013), *long* \$19.4k (Aug-2018) and short \$445.6k (Aug-2023), for a hedge market value of −\$968k (the funding leg is not modelled). The +50bps parallel P&L falls from −\$38,594 to +\$57; the individual key-rate residuals are at most \$606 (20Y), from second-order convexity.

2 Part II — Python Labs: US Treasury Yield Curve Reaction to Trump Tariff Threats

Data, event set and conventions

Data. Daily US Treasury constant-maturity yields for ten maturities, from 3 months to 30 years (FRED series `DGS3MO` to `DGS30`), downloaded with the course helper `fred.py`. The sample used in Q1 runs from 2 January 2015 to 1 August 2025, i.e. 2,647 trading days. Yields are in percent; all results are reported in basis points (bp).

Code. A single run of `ga_project/06_event_study.py` reproduces every number, table and figure of this part (output files are prefixed `pl_`). The original course notebook is in `ga_project/python_labs/`.

Event set. The notebook's `trump_actions` and `country_responses` dictionaries contain 75 events: 36 Trump actions and 39 foreign responses (Python silently keeps only the last entry of each duplicated date key, so two listed events are lost). Mapping the events to trading days:

- 8 events fall after 1 August 2025, when the FRED download ends, and are dropped;
- 16 events fall on a weekend or a market holiday and are moved to the next trading day (e.g. the events of Saturday 5 and Sunday 6 April 2025 both map to Monday 7 April);
- 9 trading days carry two events. On 8 of them a Trump action and a foreign response share the same day (e.g. 2018-07-06, 2019-08-23, 2025-03-04); on the ninth, the 4 July 2025 holiday collapses onto the 7 July action.

The remaining 67 events therefore correspond to **58 distinct event days**, each counted once. All 58 have a full pre-event window; 51 have a full 30-day post-event window, because the last events of the sample are followed by fewer than 30 trading days of data.

Methodology (as required by the assignment). Let $t_1, \dots, t_N$ be the event days and y_t a yield, or one of the three curve factors

$$\text{level} = y_{10}, \qquad \text{slope} = y_{10} - y_2, \qquad \text{curvature} = 2\,y_5 - y_2 - y_{10},$$

where y_m is the m-year yield. For each horizon $k = -30, \dots, +30$ trading days, the abnormal change is

$$\mathrm{AC}(k) = \frac{1}{N}\sum_{i=1}^{N}\left(y_{t_i+k} - y_{t_i}\right) - \frac{1}{|\mathcal{B}_k|}\sum_{s\in\mathcal{B}_k}\left(y_{s+k} - y_s\right),$$

where $\mathcal{B}_k$ is the set of non-event days s such that both s and $s+k$ lie more than 30 trading days away from every event. The first term is the average change around the events; the second is the "normal" change over the same horizon.

Reading the results. Because $\mathrm{AC}(k)$ is measured relative to the event day, a *positive* value at $k < 0$ means that yields were *higher* before the event, i.e. they *fell* into it. To avoid this sign trap, every table reports three windows as the abnormal change in yield *over the window* (positive = yields rose):

- **pre-event** $[-30, 0]$: $y_t - y_{t-30}$, i.e. $-\mathrm{AC}(-30)$;
- **announcement** $[-1, +1]$: $y_{t+1} - y_{t-1}$, i.e. $\mathrm{AC}(+1) - \mathrm{AC}(-1)$;
- **post-event** $[0, +30]$: $y_{t+30} - y_t$, i.e. $\mathrm{AC}(+30)$.

Assumptions.

(A1) Business days are the trading days present in the FRED data; weekend and holiday events move to the next trading day.

(A2) The "reaction of targeted countries" (Q1.3) is measured on the *US* curve on the dates of the foreign responses, since only US yields are provided.

(A3) Q1 uses the notebook's event list, sample and baseline unchanged; all corrections are made in Q2.1.

(A4) To judge magnitudes we also report the *raw* average change around the events with its cross-sectional standard error (s.e.). These s.e. treat events as independent; since windows overlap, they overstate precision and are only a first indication.

Q1.1 — Market reaction to Trump's announcements and actions

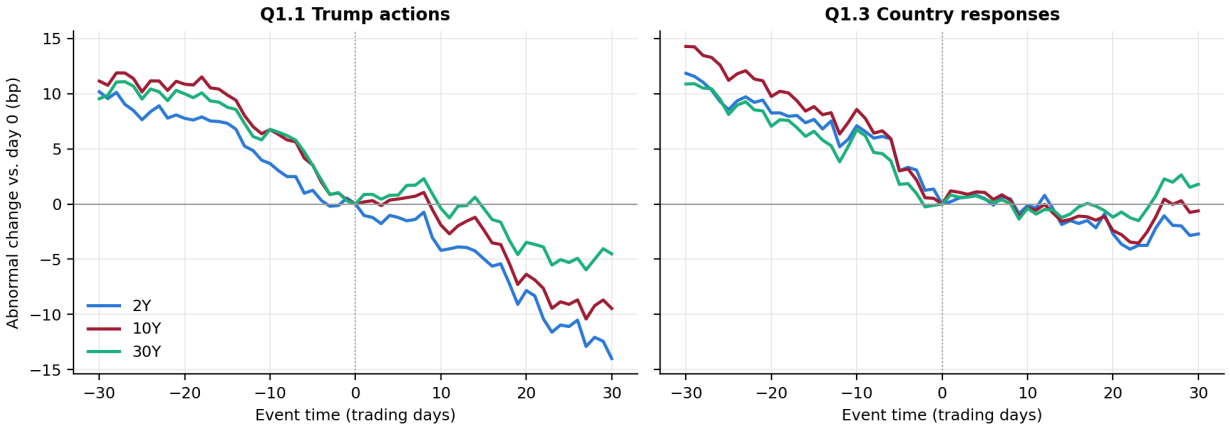


Figure 7: Abnormal change relative to the event day (bp), assignment methodology. Left: Trump actions (Q1.1). Right: foreign responses (Q1.3). A line falling from left to right means that yields fell through the window.

	3M	2Y	5Y	10Y	30Y
Panel A. Abnormal change (assignment methodology), bp					
Pre-event $[-30, 0]$	−6.4	−10.2	−11.6	−11.1	−9.5
Announcement $[-1, +1]$	−0.5	−1.6	−0.8	−0.2	+0.6
Post-event $[0, +30]$	−7.1	−14.0	−13.6	−9.5	−4.5
Panel B. Post-event window decomposed: abnormal = raw − baseline, bp					
Raw change around events	+1.4	−6.8	−7.4	−3.5	+1.0
(s.e.)	(3.1)	(5.3)	(5.5)	(5.2)	(4.8)
Non-event baseline	+8.5	+7.2	+6.2	+5.9	+5.5

Table 7: Trump actions: 32 event days in the pre-event and announcement windows, 28 in the post-event window. Panel B may not add up exactly because of rounding.

Discussion.

- **No announcement effect.** Over $[-1, +1]$ the curve barely moves: −1.6 bp on the 2Y and −0.2 bp on the 10Y, against raw s.e. of about 1.2–1.5 bp. Nothing is distinguishable from zero.
- **Large abnormal declines on both sides of the event.** Taken at face value, Panel A says that yields fall by 10–12 bp (2Y to 10Y) in the 30 days *before* the events and by a further 5–14 bp *after* them, most at the front end. The textbook reading is a negative growth shock: flight to quality into Treasuries and higher odds of Fed cuts, which lowers the 2Y most.
- **But most of it is the baseline.** Panel B shows that the raw change after the events is small and statistically zero: −3.5 bp on the 10Y (s.e. 5.2) and +1.0 bp on the 30Y. The "abnormal" declines come mainly from subtracting a non-event baseline of +5.5 to +8.5 bp per 30 days. That baseline is computed on days far from any event, i.e. mostly 2015–16 and 2020–24, a period dominated by the 2022–23 hiking cycle, when yields were rising.
- **The shape confirms it.** A reaction to news would be a jump around day 0; Figure 7 instead shows a smooth 60-day ramp that starts a month before the announcements. This is the signature of a trend difference between event periods (2018–19, 2025) and the baseline period, not of a tariff effect.

Answer. With the required methodology, Treasury yields appear to fall around Trump's tariff actions (about −10 to −14 bp at the 2Y–5Y over 30 days). However, there is no announcement effect, and the post-event decline is driven by the choice of baseline rather than by the events. Q2.1 tests this directly.

Q1.2 — Level, slope and curvature

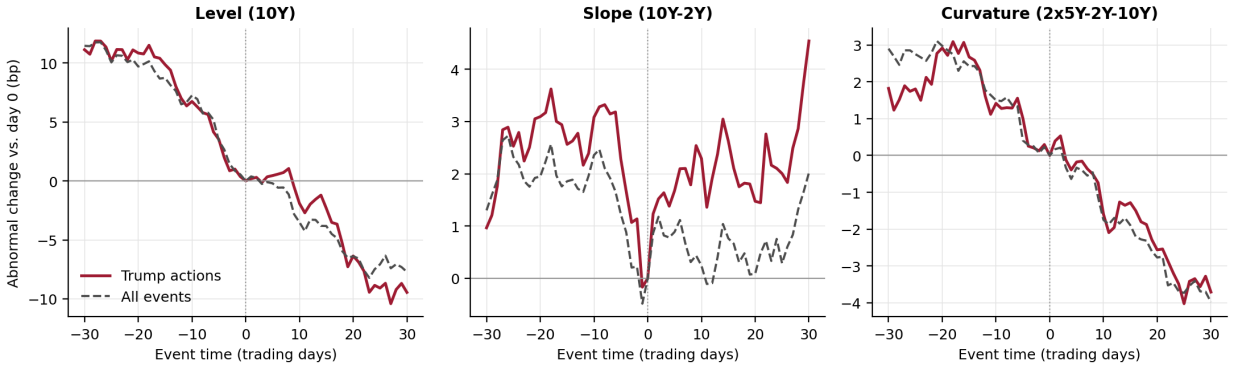


Figure 8: Abnormal change of level, slope and curvature (bp): Trump actions (red) vs. all events (dashed).

	Trump actions			All events		
Abnormal change, bp	Level	Slope	Curv.	Level	Slope	Curv.
Pre-event $[-30, 0]$	-11.1	-1.0	-1.8	-11.5	-1.3	-2.9
Announcement $[-1, +1]$	-0.2	$+1.4$	$+0.1$	-0.2	$+1.3$	-0.1
Post-event $[0, +30]$	-9.5	$+4.5$	-3.7	-7.7	$+2.0$	-4.0
Raw post-event change (s.e.)	-3.5 (5.2)	$+3.3$ (1.9)	-4.4 (1.2)			

Table 8: Level (= 10Y), slope (= 10Y − 2Y) and curvature (= 2×5Y − 2Y − 10Y).

Discussion.

- **Level** reproduces Q1.1 by construction, since it is the 10Y. The level effect is therefore subject to the same baseline caveat: raw -3.5 bp, s.e. 5.2. A level factor based on all maturities (first principal component) is used in Q2.1.
- **Slope.** After Trump actions the curve *steepens*: $+1.4$ bp over $[-1, +1]$ and $+4.5$ bp over 30 days, while the level falls. This is a *bull steepening*: the front end, anchored to expected policy rates, rallies more than the 10Y, which also carries tariff-driven inflation risk and a term premium. Unlike the level, the slope is barely affected by the baseline (-1.3 bp): the raw steepening is $+3.3$ bp (s.e. 1.9). It is partly offset by a flattening in the run-up (-1.0 bp abnormal), so the evidence is weak but this is the most event-like pattern in Q1.
- **Curvature** falls after the events (-3.7 bp): the 5Y belly rallies more than the average of the 2Y and 10Y wings. This would be consistent with cuts priced over the next 2–5 years. However, curvature also falls *before* the events (raw -2.5 bp) and after foreign responses (raw -4.5 bp). That points to a downward trend in curvature during the event periods rather than a reaction to tariff news.
- **All events vs. Trump actions.** Level and slope effects are weaker when foreign responses are included (post-event slope $+2.0$ vs. $+4.5$ bp), while curvature is unchanged, consistent with the trend interpretation.

Q1.3 — Market reaction to the targeted countries' responses

Discussion.

- **Before the responses** yields fall by 11–15 bp (2Y to 30Y). This is largely mechanical: a response typically comes a few days after the US measure it answers, so its pre-event window contains the US action itself. In eight cases the two fall on the same trading day.

	3M	2Y	5Y	10Y	30Y	Slope	Curv.
Panel A. Abnormal change (assignment methodology), bp							
Pre-event $[-30, 0]$	−2.6	−11.8	−15.3	−14.3	−10.9	−2.4	−4.4
Announcement $[-1, +1]$	+0.1	−1.1	−0.2	+0.7	+0.9	+1.8	0.0
Post-event $[0, +30]$	−3.2	−2.7	−3.7	−0.6	+1.8	+2.1	−4.1
Panel B. Raw post-event change, bp							
Raw change	+4.2	+4.1	+2.2	+4.8	+6.5	+0.6	−4.5
(s.e.)	(2.3)	(3.7)	(3.8)	(3.8)	(3.7)	(2.2)	(1.0)

Table 9: Foreign responses: 34 event days in the pre-event window, 33 in the announcement window, 31 in the post-event window.

- **At the announcement** nothing moves beyond noise (at most +0.9 bp on yields, +1.8 bp on the slope).
- **After the responses**, the abnormal change is close to zero (10Y −0.6 bp). In raw terms, however, yields *rise* by 2–7 bp, the 30Y the most (+6.5 bp, s.e. 3.7). These rises are absorbed by the rising baseline. So "no abnormal reaction" means that yields behaved like in an average non-event month, not that they were flat.
- **Economic reading.** Retaliation has two opposite effects on US yields. It hurts US exporters, which is growth-negative and lowers yields. It also raises import prices and threatens foreign demand for Treasuries, which pushes up inflation expectations and the term premium. The mild rise of the long end relative to the front end (slope +1.8 bp at the announcement) suggests that the second channel slightly dominates. The evidence, however, is weak and concentrated in 2025 (Q1.4).

Q1.4 — First vs. second mandate

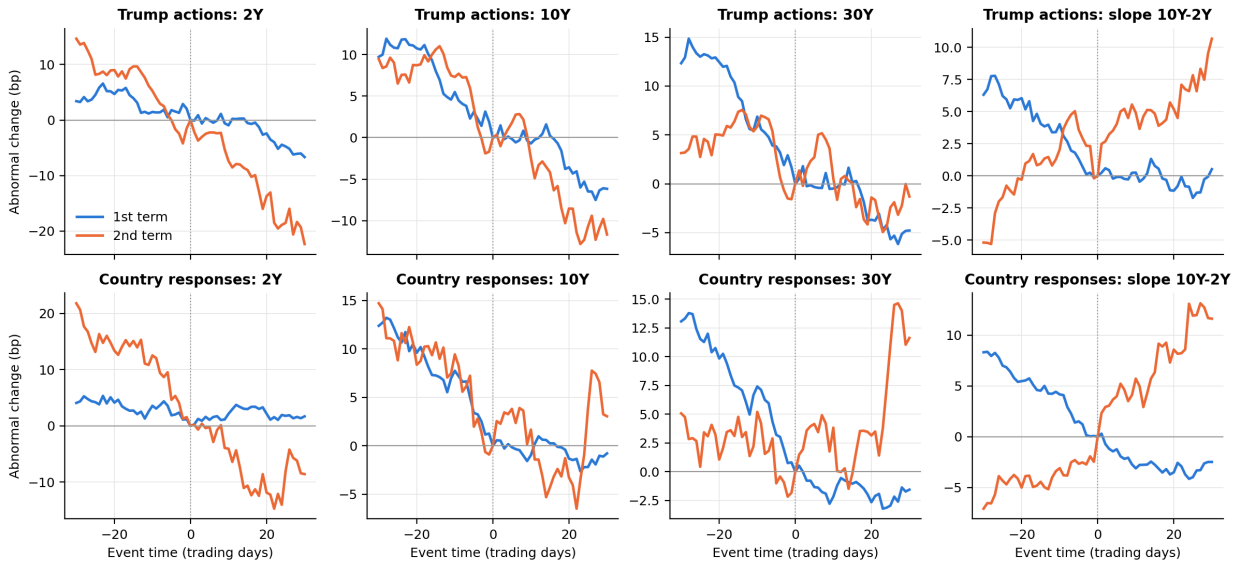


Figure 9: Abnormal change by mandate: first term (2017–20, blue) vs. second term (2025, orange). Top: Trump actions. Bottom: foreign responses.

Discussion. The two mandates look very different, but for a different reason than Panel A alone suggests.

- **First term (2018–19): no measurable reaction.** After Trump actions the abnormal changes suggest a parallel rally of 5–7 bp. In raw terms, however, the 2Y, 10Y and 30Y are all within 1 bp of zero (s.e. about 6–7 bp): the apparent rally is entirely the baseline. The only clear

		Pre-event $[-30, 0]$, bp				Post-event $[0, +30]$, bp			
Group	n	2Y	10Y	30Y	Slope	2Y	10Y	30Y	Slope
Abnormal change (assignment methodology)									
Actions, 1st term	18/18	−3.4	−9.7	−12.3	−6.3	−6.7	−6.2	−4.8	+0.5
Actions, 2nd term	14/10	−14.7	−9.5	−3.1	+5.2	−22.4	−11.7	−1.3	+10.7
Responses, 1st term	22/22	−4.1	−12.4	−13.1	−8.3	+1.7	−0.8	−1.6	−2.5
Responses, 2nd term	12/9	−21.8	−14.7	−5.1	+7.1	−8.6	+3.0	+11.6	+11.6
Raw post-event change (s.e.)									
Actions, 1st term						−0.7 (6.7)	−0.7 (6.8)	+0.8 (6.1)	0.0 (2.2)
Actions, 2nd term						−17.9 (7.9)	−8.7 (8.5)	+1.2 (8.2)	+9.2 (3.0)
Responses, 2nd term						−4.1 (7.8)	+6.0 (7.4)	+14.1 (7.4)	+10.1 (4.5)

Table 10: Yield change by mandate. Actions = Trump actions; n = event days in the pre-/post-event windows.

first-term feature is a *flattening into the events* (abnormal slope −6.3 bp after actions, −8.3 bp after responses). This matches the general flattening of the 2018 hiking cycle more than tariff news. Tariffs were narrower (mainly China, plus steel and aluminium), inflation was low, and the Fed's reaction function dominated the curve.

- **Second term (2025): a steepening.** Here the raw numbers move:
  - After Trump actions, the 2Y falls by 18 bp while the 30Y does not move, giving a *bull steepening* (slope +9.2 bp, s.e. 3.0): the front end prices Fed cuts against weaker growth.
  - After foreign responses, the 30Y *rises* by 14 bp (slope +10.1 bp), a *bear steepening*.

  The long end no longer acted as a safe haven. Universal tariffs are inflationary, fiscal concerns were acute (Moody's downgrade in May, the One Big Beautiful Bill Act in July), and markets questioned foreign demand for Treasuries during the April "sell America" episode. The shock was absorbed by the term premium rather than by lower long yields.
- **Reliability.** The second-term samples are small: 10 and 9 event days in the post-event window. They are also concentrated between February and July 2025, so their 30-day windows overlap and represent only a handful of independent episodes. The s.e. above are therefore too small, and the differences between mandates are indicative only. Q2.1 tests which of them survive proper inference.

Q2.1 — Limitations, improvements, and how they change Q1

(a) Limitations of the design

1. **Biased baseline (regime effect).** The "normal" change is estimated on non-event days, which fall mostly in 2015–16 and 2020–24, a period dominated by the 2022–23 hiking cycle. The 10Y non-event baseline therefore rises by +6.5 bp per 30 days, while the average 30-day change was −3.6 bp in 2017–20 and −5.8 bp in 2025 (Table 11). Subtracting a rising baseline from falling-rate periods mechanically creates a negative "abnormal" change. Q1 shows that this explains most of the post-event declines (raw vs. abnormal changes in Tables 7 and 9).
2. **Overlapping windows.** Events come in bursts (e.g. 14 events between 2025-02-01 and 2025-04-10). The 30-day windows therefore overlap and contain each other's events: observations are not independent, and each long window mixes many news items.
3. **No valid inference.** The assignment's design produces averages without a test. The naive standard errors added in Q1 treat overlapping windows as independent, so they overstate precision. With 61 horizons and several series, some "large" numbers are bound to appear by chance.
4. **Event-list quality.**
   - Two entries are silently lost to duplicated dictionary keys (2025-08-06 and 2025-03-04).

- Some entries are not shocks: "no retaliation" announcements, a status quo, and a US court ruling listed as a foreign response.
- Escalations and de-escalations (Phase One deal, delays, exemptions) are averaged together, so their effects offset each other.
- Major milestones are missing: the 90-day pause (2025-04-09), the US–China Geneva agreement (2025-05-12) and the US–EU deal (2025-07-27).
- Some descriptions are inaccurate (145% was the US rate on China).

5. **Sample truncation.** The download stops on 2025-08-01: eight August-2025 events are dropped and late-July windows are cut short.
6. **Timing.** Several announcements came after the market close ("Liberation Day" on 2025-04-02 was announced after 4pm ET) or on weekends, so day 0 is not always the reaction day (Figure 11).
7. **Confounding news.** Some event days coincide with FOMC decisions, and Trump actions and foreign responses share the same trading day eight times in the notebook list.
8. **Measurement.** Level = 10Y duplicates Q1.1. Constant-maturity par yields are used rather than zero rates, and daily closes cannot isolate an announcement from other news on the same day.

Period	2015–16	2017–20 (1st term)	2021–24	2025 (2nd term)
Mean 30-day 10Y change (bp)	+3.3	−3.6	+9.7	−5.8

Table 11: Average 30-day change of the 10Y yield by period. The non-event baseline of the assignment (+6.5 bp) is dominated by the 2021–24 hiking cycle.

(b) Improvements implemented

All improvements are in `06_event_study.py`, function `part_q2`.

1. **Cleaned and signed event list.**
   - Duplicated keys merged and six non-events removed (list in the code).
   - Each event signed as escalation (+1) or de-escalation (−1).
   - Three missing de-escalation milestones added.
   - Result: 72 events, of which 63 escalations and 9 de-escalations.
2. **Extended sample** to 2025-10-31, so that every event has a full window.
3. **Short windows** $[-1, +1]$ and $[0, +5]$ to capture the announcement effect. The $[-1, +1]$ window also covers after-close announcements, whose reaction occurs on day +1.
4. **Baseline from the same mandate.** The expected change is the average change over windows of the same length that start on ordinary days of the same mandate, i.e. days with no event within ±1 day (913 days in the first term, 127 in the second). Each event is thus compared with a typical window of its own rate regime, not with a 2015–2025 average.
5. **Placebo inference.** For each group we draw 5,000 sets of pseudo-event dates from the same pools, keeping the number of events per mandate. The two-sided p-value is the share of placebo averages at least as extreme as the actual one.
   - A stricter pool (no event within ±5 days) leaves only 56 days in 2025. These are unusually calm days, so the placebo distribution becomes too narrow and the p-values too optimistic.
   - With that pool, 6 results lose significance at 5% and 5 gain it, but none of the conclusions below changes (`pl_q2_placebo_sensitivity.csv`).

6. **De-clustering.** The 30-day window uses only the first event of each cluster (no other event in the previous 5 trading days).
7. **Robustness.** We exclude events with a same-day foreign response and events within ±1 day of an FOMC decision, and we replace the 10Y by a PCA level (first principal component of daily changes). For 2025 we also drop April, the month of "Liberation Day" and of the Treasury basis-trade deleveraging (Table 13).

	$[-1,+1]$				$[0,+5]$			$[0,+30]$, cluster heads		
Group	2Y	10Y	30Y	Slope	10Y	30Y	Slope	2Y	10Y	n
Trump escalations	+1.2	+1.5	+2.1	+0.3	+1.8	+2.0	+2.0*	+6.6	+6.1	31/18
1st term	−2.3	−2.2	−1.5	+0.1	−0.6	−0.5	−0.4	+7.9	+7.2	16/12
2nd term	+5.1**	+5.5**	+6.0**	+0.4	+4.3	+4.7*	+4.6**	+4.1	+4.1	15/6
Foreign escalations	+1.8	+3.5**	+3.7**	+1.7*	+4.3**	+4.1**	+2.1*	+4.6	+7.7	32/10
1st term	+0.2	+0.4	+0.7	+0.2	+0.6	−0.4	−1.4	+9.5	+7.7	20/7
2nd term	+4.6**	+8.8**	+8.8**	+4.2**	+10.5**	+11.6**	+7.7**	−6.9	+7.6	12/3
De-escalations (all)	+7.0**	+6.7**	+5.9**	−0.3	+1.9	+2.9	+2.6	−20.6**	−19.7**	9/4

Table 12: Improved design: abnormal change in bp (positive = yields up) relative to ordinary windows of the same mandate, extended sample, cleaned events. Placebo p-values: $^{*}p < 0.10$, $^{**}p < 0.05$. n = events in the short / 30-day windows.

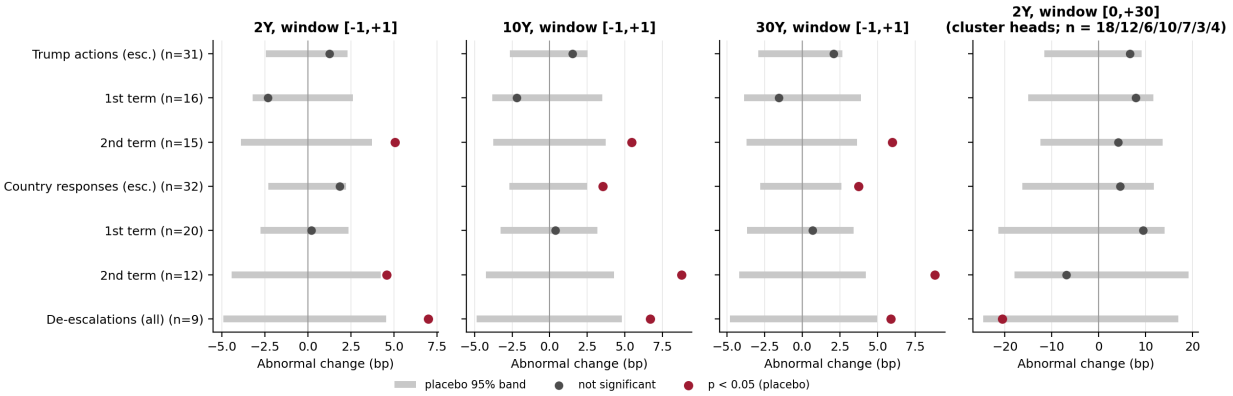


Figure 10: Improved estimates (dots) against the 95% range of the placebo distribution (grey bars). Red dots are significant at 5%.

		10Y		30Y	
2025 escalations, window $[-1,+1]$	n	bp	p	bp	p
Trump actions, all	15	+5.5	0.005	+6.0	0.002
excl. April 2025	13	+5.7	0.006	+5.7	0.005
excl. same-day foreign response	12	+3.4	0.112	+3.7	0.072
Foreign responses, all	12	+8.8	< 0.001	+8.8	< 0.001
excl. April 2025	9	+6.7	0.005	+6.7	0.006
excl. same-day Trump action	9	+7.1	0.004	+6.8	0.005

Table 13: Robustness of the 2025 announcement effect (placebo p-values). No 2025 escalation falls within ±1 day of an FOMC decision, so that exclusion changes nothing.

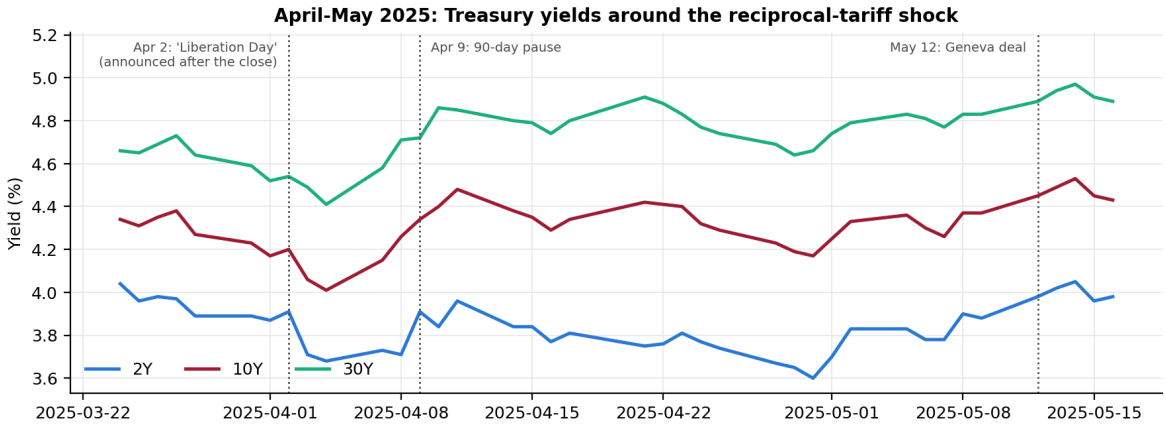


Figure 11: Daily yields around "Liberation Day". The announcement came after the close on 2 April, so the reaction is on 3 April (−20 bp 2Y, −14 bp 10Y). From 7 to 10 April the long end sold off (+45 bp on the 30Y between the 4 and 10 April closes) while tariffs were still escalating.

	Assignment design (Q1)	Improved design (Q2)
Q1.1	Abnormal declines of 10–14 bp after Trump actions, but no announcement effect and a raw change close to zero.	**Confirmed: no effect when both mandates are pooled.** Trump escalations have no significant announcement effect ($p \geq 0.13$; 1 of 20 robustness cells below 5%). The 30-day decline disappears (+6.6 bp on the 2Y, n.s.) once each event is compared with its own rate regime.
Q1.2	Level down, slope up (+4.5 bp): an apparent bull steepening.	No level effect when pooled. A steepening over $[0, +5]$ (+2.0 bp, $p = 0.07$) comes entirely from 2025 (+4.6 bp**), and it follows a *rise* in yields, not a front-end rally: a bear steepening.
Q1.3	Abnormal change near zero after responses, while raw 10Y and 30Y yields rise by 5–7 bp.	**Significant rise in long yields**: 10Y +3.5** and 30Y +3.7** at the announcement, +4.1 to +4.3** over five days. The effect is entirely due to 2025 (10Y and 30Y +8.8**, slope +4.2**) and survives dropping April.
Q1.4	1st term: no raw reaction. 2nd term: front-end decline and steepening.	**1st term: no effect at any horizon. 2nd term: yields rise** by 5–6 bp across the curve on Trump escalation days and by 9 bp at the long end after retaliation; the curve then steepens. The 30-day front-end decline of 2025 is not event-specific: it is the general 2025 trend. Treasuries did not act as a safe haven.
New	—	**De-escalations also raise yields** (+6 to +7 bp, with an unchanged slope; e.g. +20 bp on the 2Y on 9 April 2025). This is a "risk-on" repricing of expected Fed cuts, which pooling escalations with de-escalations had hidden.

Table 14: Conclusions before and after the methodological improvements.

(c) How the improvements change our answers to Q1

Overall. The smooth 60-day patterns of Q1 are mostly a rate-regime effect plus clustering, not tariff news. Once these are removed, three findings remain:

1. First-term tariff news had no measurable effect on Treasury yields.
2. In 2025, escalation days, and above all foreign retaliation, *raised* yields and steepened the curve. The rise is largest at the long end, consistent with a rising term premium (inflation, fiscal and foreign-demand concerns) rather than a flight to safety. It is not driven by April 2025 alone.
3. De-escalations raised yields roughly in parallel, as markets priced out Fed cuts. The *slope* separates the two channels: escalations steepen the curve, de-escalations do not.

Remaining limitations.

- **Multiple testing.** Table 12 reports 84 tests, 21 of them significant at 5%, against about 4 expected by chance. The significant cells, however, are concentrated in the 2025 groups, whose events fall in the same few months, so they are far from 21 independent findings.
- **Small samples.** The second-term groups have 12–15 events in the short windows and only 3–6 in the 30-day windows. The 30-day results, including the -20 bp for de-escalations ($n = 4$), are anecdotal.
- **Overlaps remain.** Removing same-day foreign responses weakens the 2025 Trump effect ($p = 0.07$–0.11), so part of it may be retaliation news released on the same day.
- **Daily data.** A cleaner design would use intraday Treasury futures in 30-minute windows around each announcement, and zero-coupon rather than par yields.
- **Foreign curves.** With the relevant data, foreign yields could measure the reaction in the targeted countries directly.